

Comparative Analysis of Digital Advertising Performance Across Google, Meta, and TikTok

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Abstract

Digital advertising platforms are commonly evaluated using metrics such as click-through rate, cost per click, and return on ad spend. However, these measures do not fully capture the multidimensional nature of campaign performance. This study analyzes a simulated dataset of 1,800 campaigns across Google Ads, Meta Ads, and TikTok Ads using descriptive statistics, Multiple Factor Analysis (MFA), MANOVA, clustering, regression, optimization, and machine learning.

The results suggest systematic differences across platforms within the dataset. TikTok Ads exhibits stronger engagement and return metrics, Google Ads appears more cost-intensive, and Meta Ads falls between the two. MFA provides a reduced-dimensional representation of campaign performance, and MANOVA results indicate that platform differences remain significant after accounting for campaign type, industry, and country. Regression results show a nonlinear relationship between ad spend and revenue with diminishing returns. Optimization demonstrates that marginal-return-based allocation outperforms equal allocation. Because the data are simulated, these findings should be interpreted as demonstrating a statistical framework rather than definitive real-world conclusions.

1 Introduction

Digital advertising has become a central component of modern marketing strategy. Platforms such as Google Ads, Meta Ads, and TikTok Ads allow firms to deliver targeted campaigns, monitor engagement in real time, and evaluate performance using a wide range of metrics, including impressions, clicks, click-through rate (CTR), cost per click (CPC), conversions, cost per acquisition (CPA), revenue, and return on ad spend (ROAS). Prior research highlights that social media platforms can enhance firm value and competitiveness by

enabling more direct engagement with consumers (Ahmad et al. 2018).

Despite the availability of detailed performance metrics, comparing advertising platforms is not straightforward. These metrics are inherently interrelated: clicks arise from impressions, CTR depends on the relationship between clicks and impressions, and financial outcomes such as revenue and ROAS depend on both conversions and spending. As a result, evaluating platforms based on a single metric can produce incomplete or misleading conclusions. A platform that performs well in terms of reach may differ in efficiency or financial return, making it necessary to consider campaign performance from a multivariate perspective.

In addition to platform-level differences, advertising outcomes are influenced by campaign-specific factors such as campaign type, industry, and geographic context. These factors can affect user engagement, bidding behavior, and conversion rates, further complicating direct comparisons. From a strategic perspective, marketing efforts must be carefully integrated across platforms. Prior work emphasizes that firms must align digital campaigns with broader marketing objectives and use appropriate tools effectively (Bekoglu and Onaylı 2016). This reinforces the importance of analyzing platform performance within a broader campaign context rather than in isolation.

The purpose of this study is to examine whether campaign performance differs systematically across Google Ads, Meta Ads, and TikTok Ads, and to evaluate how those differences can inform budget allocation decisions. The analysis proceeds in several stages. First, descriptive statistics provide an overview of platform and campaign-type differences. Second, Multiple Factor Analysis (MFA) is used to summarize the joint structure of advertising metrics in a lower-dimensional space. Third, multivariate analysis of variance (MANOVA) is used to test whether platform differences remain significant after accounting for campaign characteristics. Fourth, clustering is used to identify recurring campaign performance profiles. Fifth, regression analysis is used to examine the relationship between ad spend and revenue, with particular attention to diminishing returns. Finally, optimization methods and predictive models are used to translate these findings into actionable decision-making strategies.

This study addresses two related questions. First, do digital advertising platforms exhibit systematic differences in campaign performance after accounting for campaign context? Second, how can those differences be used to guide budget allocation when returns to advertising spend may be nonlinear? By combining descriptive analysis, multivariate methods, regression modeling, clustering, and optimization, this study develops a unified framework for understanding and improving digital advertising performance.

Because the dataset used in this study is simulated, the findings should be interpreted as illustrating a statistical framework rather than providing definitive conclusions about real-world platform performance. Nonetheless, the approach demonstrates how multiple analytical techniques can be integrated to support data-driven decision-making in digital marketing.

2 Data

The dataset contains 1,800 campaign-level observations across Google Ads, Meta Ads, and TikTok Ads. Variables include platform, campaign type, industry, country, and performance metrics such as impressions, clicks, CTR, CPC, ad spend, conversions, CPA, revenue, and ROAS.

Variables are grouped into three conceptual categories:

- Exposure: impressions, clicks
- Efficiency: CTR, CPC, CPA, ROAS
- Financial outcomes: ad spend, revenue, conversions

Because the dataset is simulated, the results reflect patterns within the generated data rather than externally validated platform performance.

3 Methods

Several modeling considerations are important for interpreting the results. First, many advertising variables in the dataset are strongly related to one another. For example, clicks, conversions, revenue, and return on ad spend are not independent quantities, and modeling them directly in the same framework can introduce multicollinearity. To address this issue, dimension reduction is used before conducting multivariate testing.

Second, the analysis is associational rather than causal. Differences across platforms should not be interpreted as causal platform effects because campaign outcomes may also reflect unobserved factors such as ad quality, targeting strategy, audience composition, or bidding behavior. The goal of this study is therefore to identify systematic patterns in performance rather than estimate causal effects.

Third, because the dataset is simulated, the results should be interpreted as demonstrating a statistical framework rather than providing externally valid real-world estimates. The simulated data are designed to reflect realistic relationships, but the underlying data-generating process is not fully specified.

3.1 Descriptive Comparison

The first stage of the analysis focuses on descriptive comparison. Mean values of CTR, CPC, CPA, and ROAS are computed by platform to summarize differences in engagement efficiency, cost, and return. Mean values of ROAS, CPA, conversions, and ad spend are also computed by campaign type to assess whether campaign objectives are associated with systematic variation in outcomes.

These summaries provide an interpretable starting point for the analysis, translating the dataset into metrics that advertisers commonly use in practice.

3.2 Multiple Factor Analysis

Because the advertising metrics are highly correlated, Multiple Factor Analysis (MFA) is used to summarize their joint structure. MFA is particularly appropriate when variables can be grouped into meaningful blocks, as is the case here with exposure, efficiency, and financial outcomes.

The MFA transformation can be written as:

$$Z = XW,$$

where X is the standardized data matrix and W is a weighting matrix that ensures each variable group contributes equally to the analysis. This weighting is important because the groups differ in size, and without it larger groups could dominate the results.

In practical terms, MFA produces dimensions that summarize patterns across campaigns. These dimensions can be interpreted using variable loadings and visualized to understand how campaigns and platforms differ in overall performance.

3.3 MANOVA for Platform Differences

After dimension reduction, multivariate analysis of variance (MANOVA) is used to test whether platforms differ in the reduced-dimensional space. This is a MANOVA rather than a one-way ANOVA because the response consists of multiple variables simultaneously, specifically the first two MFA dimensions.

The test is based on Wilks' Lambda:

$$\Lambda = \frac{|E|}{|E + H|},$$

where E represents within-group variation and H represents between-group variation. Smaller values of Λ indicate greater separation among group means.

The analysis uses the first two MFA dimensions because they provide the most interpretable low-dimensional representation of the data and are used for visualization. Although these two dimensions explain approximately 21.4% of the total variation, they capture the dominant patterns that distinguish campaigns in a two-dimensional space. This choice prioritizes interpretability over completeness and represents a limitation of the analysis.

Three MANOVA models are estimated. The first includes only platform as a predictor. The second adds campaign type, and the third includes industry and country. This sequence allows assessment of whether platform differences persist after accounting for additional observed factors.

3.4 Clustering Analysis

To identify recurring campaign performance profiles, k-means clustering is applied to the first two MFA dimensions. These dimensions provide a compact summary of the joint structure of the data and are therefore suitable for clustering.

The number of clusters is evaluated using silhouette width, which measures how well each observation fits within its assigned cluster relative to neighboring clusters. A three-cluster solution is selected because it provides a balance between separation and interpretability.

Clustering complements MANOVA by identifying patterns that are not tied directly to platform labels, instead revealing common performance regimes across campaigns.

3.5 Spend–Revenue Regression

The relationship between ad spend and revenue is examined using a quadratic regression model on the log-transformed scale:

$$\log(\text{Revenue} + 1) = \beta_0 + \beta_1 \log(\text{Ad Spend} + 1) + \beta_2 [\log(\text{Ad Spend} + 1)]^2 + \varepsilon.$$

A quadratic specification is used because it provides a simple and interpretable way to model diminishing returns. In advertising contexts, revenue is expected to increase with spending, but the marginal gain from additional spending may decrease due to audience saturation or reduced responsiveness.

A negative coefficient on the squared term indicates diminishing marginal returns, meaning that each additional increase in spending generates a smaller incremental increase in revenue.

3.6 Budget Optimization

The regression results motivate a budget allocation problem. Let B denote the total budget and s_G , s_M , and s_T denote spending on Google Ads, Meta Ads, and TikTok Ads, respectively. The objective is to maximize total expected revenue:

$$\max_{s_G, s_M, s_T} R_G(s_G) + R_M(s_M) + R_T(s_T)$$

subject to:

$$s_G + s_M + s_T \leq B, \quad s_G, s_M, s_T \geq 0.$$

Because the estimated spend–revenue relationship exhibits diminishing returns, the optimal solution balances marginal gains across platforms rather than allocating all budget to a single option.

In practical terms, the optimization is implemented using a stepwise greedy algorithm based on marginal returns. Instead of allocating the entire budget at once, the algorithm assigns budget incrementally. At each step, additional budget is allocated to the platform that produces the largest marginal increase in predicted revenue.

As spending increases on a given platform, its marginal return decreases due to diminishing returns. This causes the algorithm to shift allocation toward other platforms with higher remaining marginal gains. The process continues until the full budget is allocated, resulting in an allocation that approximately equalizes marginal returns across platforms.

3.7 Platform and Campaign-Type Interactions

To examine whether platform effectiveness depends on campaign objectives, interaction models are estimated involving platform and campaign type. This allows the effect of platform to vary across Search, Video, Shopping, and Display campaigns rather than assuming a constant platform effect.

3.8 Joint Optimization by Platform and Campaign Type

The optimization framework is extended by allowing spending decisions to vary jointly by platform and campaign type. Separate spend–revenue curves are estimated for each platform–campaign-type combination, and the budget is allocated across all combinations.

This extension reflects the possibility that platform performance depends on campaign objectives and allows for more targeted allocation decisions.

3.9 Predictive Modeling

Finally, campaign performance is examined from a predictive perspective. The outcome variable is defined as $\log(\text{conversions} + 1)$ to stabilize variance.

Two models are fitted using an 80/20 train–test split: a linear regression model and a random forest model. Random forests are used because they can capture nonlinear relationships and interactions without requiring explicit specification.

Model performance is evaluated using root mean squared error (RMSE) on the test set. This comparison assesses whether flexible machine learning models improve predictive accuracy relative to a simpler linear benchmark.

3.10 Rationale for Method Selection

Each method in the analysis serves a specific purpose in addressing the research questions. Multiple Factor Analysis (MFA) is used to handle the strong correlations among advertising metrics by reducing them to a smaller number of interpretable dimensions. MANOVA is then used to test whether platform differences persist across these dimensions rather than in isolated variables.

Regression analysis is used to examine the relationship between ad spend and revenue and to detect diminishing returns. The optimization framework builds directly on this relationship by translating estimated marginal returns into budget allocation decisions. Finally, random forest models are used to capture nonlinear relationships and interactions that may not be well represented in a linear model.

Together, these methods provide a unified framework that moves from description to inference and ultimately to decision-making.

4 Results

4.1 Platform-Level Performance Differences

Table 1: Average performance metrics by platform

Platform	CTR	CPC	CPA	ROAS
Google Ads	0.040	2.149	64.065	4.113
Meta Ads	0.025	1.316	39.097	6.916
TikTok Ads	0.055	1.010	29.197	9.539

Platform differences are substantial within this simulated dataset. TikTok Ads has the highest average CTR and ROAS and the lowest CPA, while Google Ads has the highest CPC and CPA. Meta Ads falls between the two. These patterns indicate that the platforms differ not only in individual metrics, but also in how they balance engagement, cost, and financial return. In particular, TikTok Ads appears to combine higher engagement with stronger return, whereas Google Ads appears more cost-intensive. Because the dataset is simulated, these results should be interpreted as patterns within the data rather than definitive real-world conclusions.

4.2 Campaign Type Effects on Performance

Table 2: Average performance metrics by campaign type

Campaign Type	ROAS	CPA	Conversions	Ad Spend
Display	6.45	45.85	183.02	6296.99
Search	7.00	44.30	190.02	6012.59
Shopping	5.98	48.00	179.14	6262.97
Video	6.34	48.36	173.75	6132.58

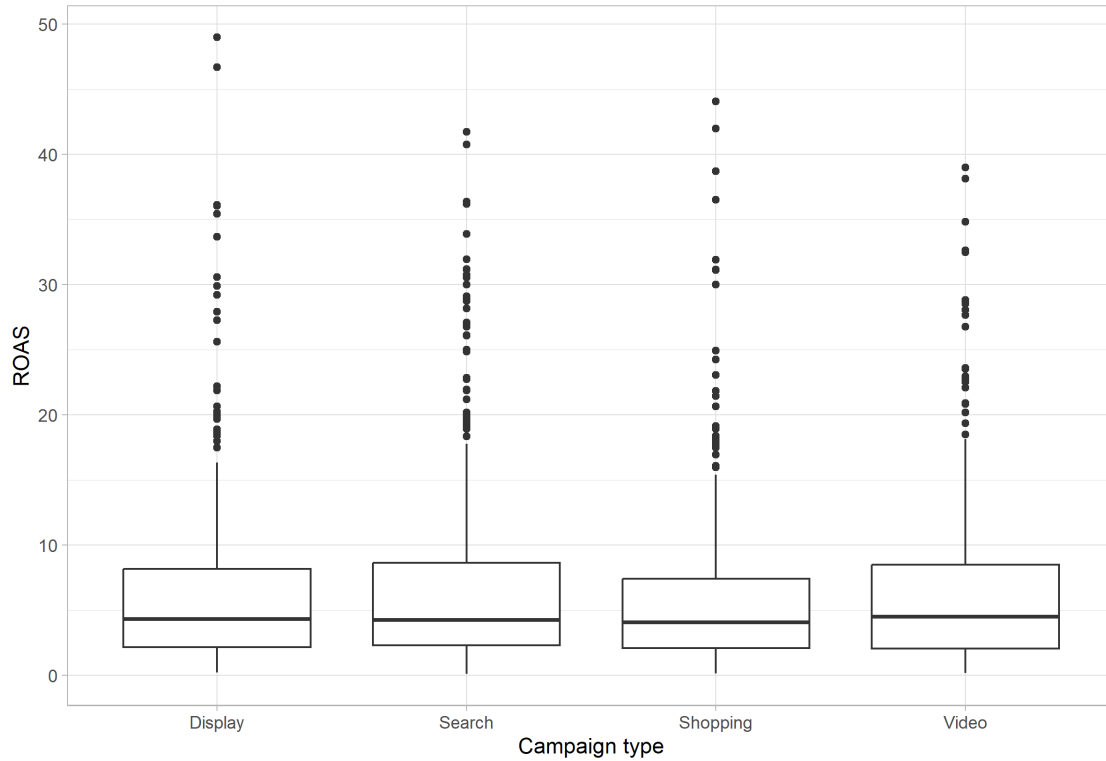


Figure 1: Distribution of ROAS by campaign type

Campaign type shows more modest differences relative to platform. Search campaigns have the highest average ROAS, while Shopping campaigns have the lowest. However, Figure 1 shows substantial overlap across campaign types, and the MANOVA results are not statistically significant. This indicates that campaign type alone does not create strong multivariate separation, suggesting that campaign performance is not sharply divided by objective. Instead, campaign outcomes appear to vary more continuously across types.

4.3 Multivariate Platform Separation

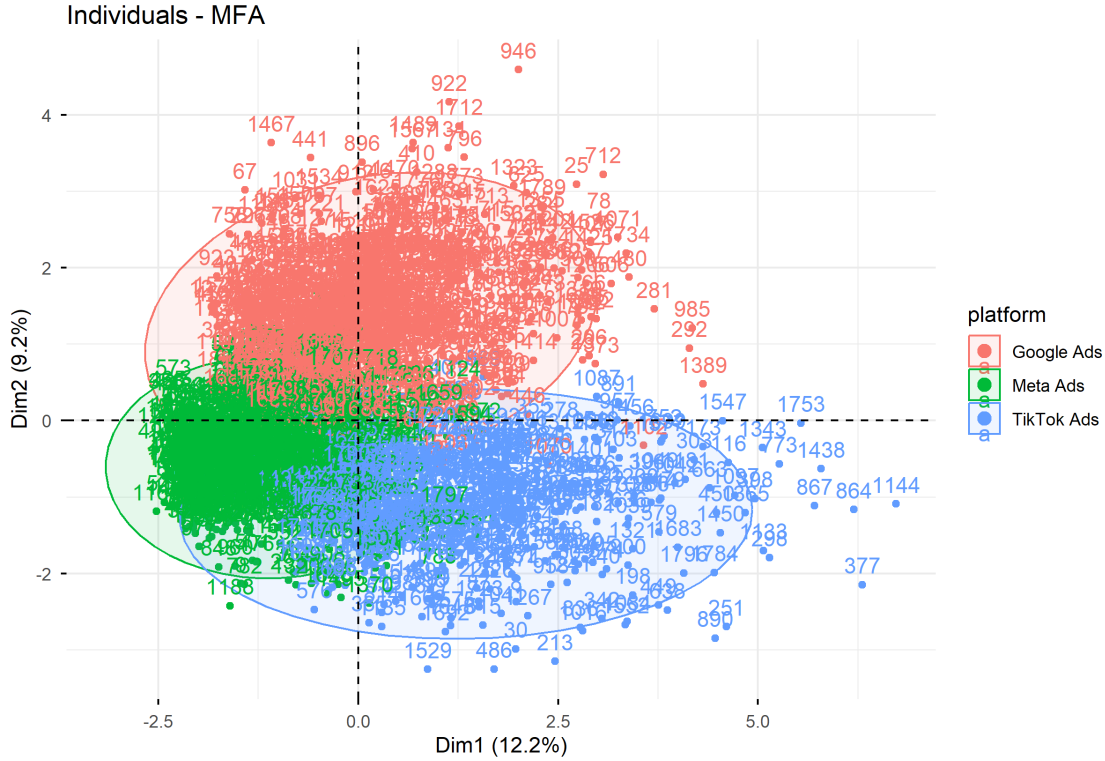


Figure 2: Campaigns in the first two MFA dimensions, colored by platform

The first two MFA dimensions explain approximately 12.2% and 9.2% of total variation. Although this does not capture all variability, it provides a useful representation of the dominant patterns. Figure 2 shows clear separation among platforms, indicating that platform differences are not limited to individual variables but instead reflect broader structural differences across multiple performance dimensions. This indicates that platform differences are not confined to individual metrics, but instead reflect broader structural differences across multiple dimensions of campaign performance. Dimension 1 is primarily associated with overall campaign scale, while Dimension 2 reflects a contrast between cost and return.

4.4 Adjusted Platform Effects

The MANOVA results show that platform differences remain highly significant after adjusting for campaign type, industry, and country. This indicates that platform is the strongest observed factor shaping performance in the reduced-dimensional space. In other words, the separation observed in the MFA plot is not fully explained by other observed campaign characteristics. Because the analysis is non-causal and based on simulated data, this result should be interpreted as evidence of systematic separation rather than a causal

platform effect.

4.5 Campaign Performance Clusters

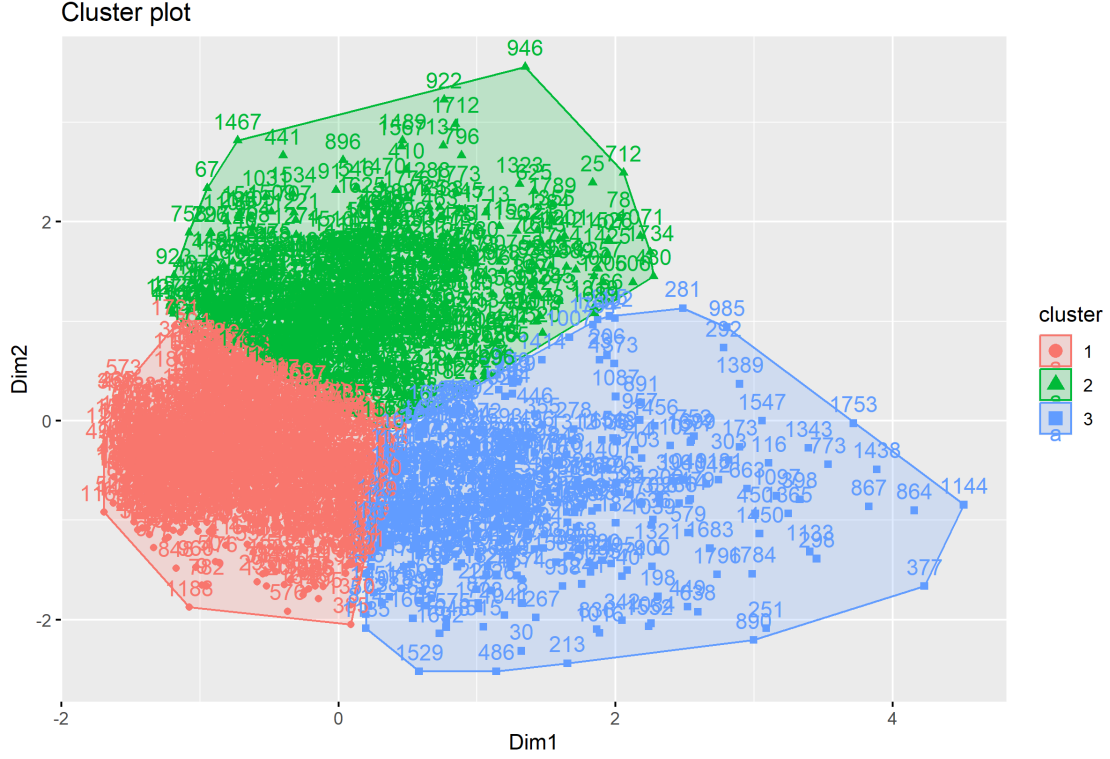


Figure 3: K-means clustering of campaigns in the first two MFA dimensions

Clustering analysis identifies three broad campaign performance profiles. The average silhouette width is approximately 0.42, indicating moderate but meaningful structure. This suggests that campaign outcomes tend to cluster into a small number of recurring performance regimes rather than being uniformly distributed.

These clusters can be interpreted as distinct performance archetypes. One cluster is characterized by high spending with relatively lower return, representing campaigns that achieve scale but at higher cost. A second cluster reflects more balanced campaigns, with moderate spending and moderate return. A third cluster represents more efficient campaigns, characterized by stronger return relative to spending.

These archetypes provide a more interpretable summary of campaign behavior and help explain how different combinations of scale, efficiency, and return emerge in practice. They also align with the MFA dimensions, where campaigns are separated based on overall scale and the trade-off between cost and return.

4.6 Diminishing Returns in Advertising Spend

Table 3: Quadratic regression coefficients for log revenue

Term	Estimate	Std. error	Statistic	p-value
Intercept	-0.8682	0.9022	-0.9623	0.336
$\log(\text{ad spend} + 1)$	1.8546	0.2282	8.1256	0.000
$[\log(\text{ad spend} + 1)]^2$	-0.0687	0.0143	-4.8050	0.000

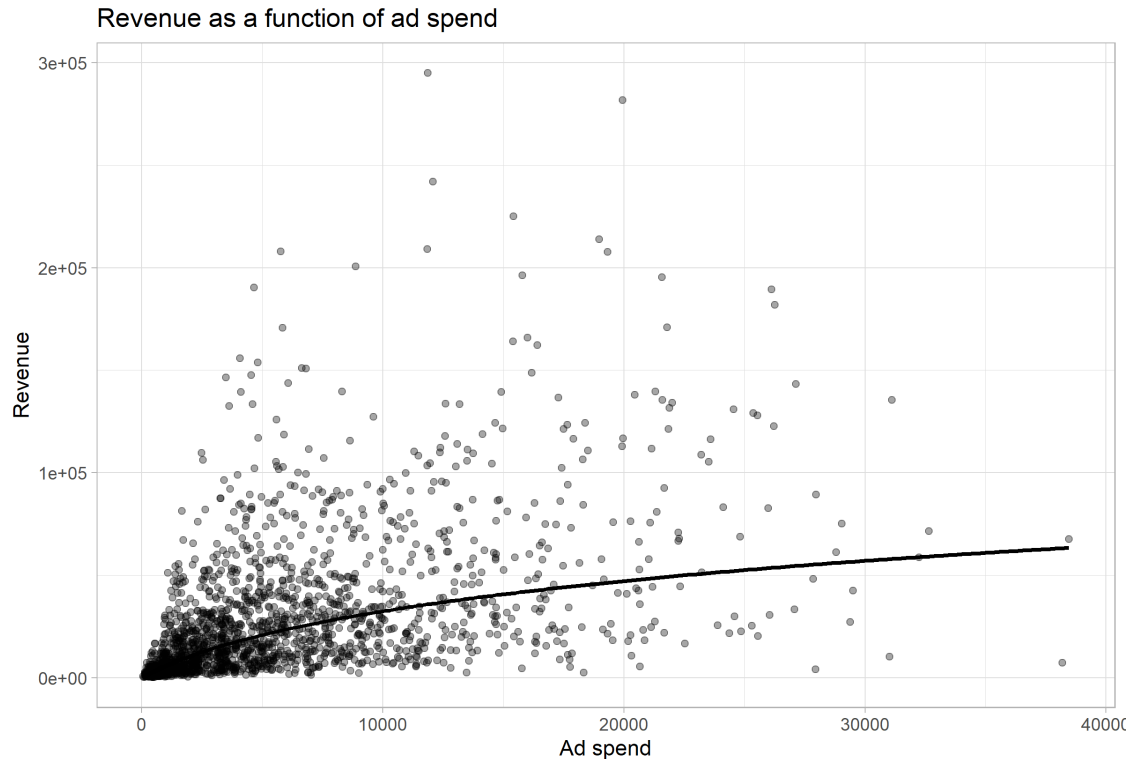


Figure 4: Observed revenue and fitted quadratic spend–revenue curve

The regression results indicate a positive but nonlinear relationship between ad spend and revenue. The squared term is negative and statistically significant, providing clear evidence of diminishing returns. Figure 4 shows that revenue increases rapidly at lower spending levels but flattens as spending grows. This pattern implies that each additional dollar of ad spend generates a smaller incremental increase in revenue. The model explains a moderate portion of the variation in revenue ($R^2 = 0.4389$), suggesting that ad spend is an important but not exclusive driver of campaign outcomes.

4.7 Platform-Level Budget Optimization

Table 4: Equal and optimized budget allocations by platform

Platform	Equal spend	Optimal spend	Optimal share
Google Ads	10000	5651.94	0.19
Meta Ads	10000	6634.13	0.22
TikTok Ads	10000	17713.93	0.59

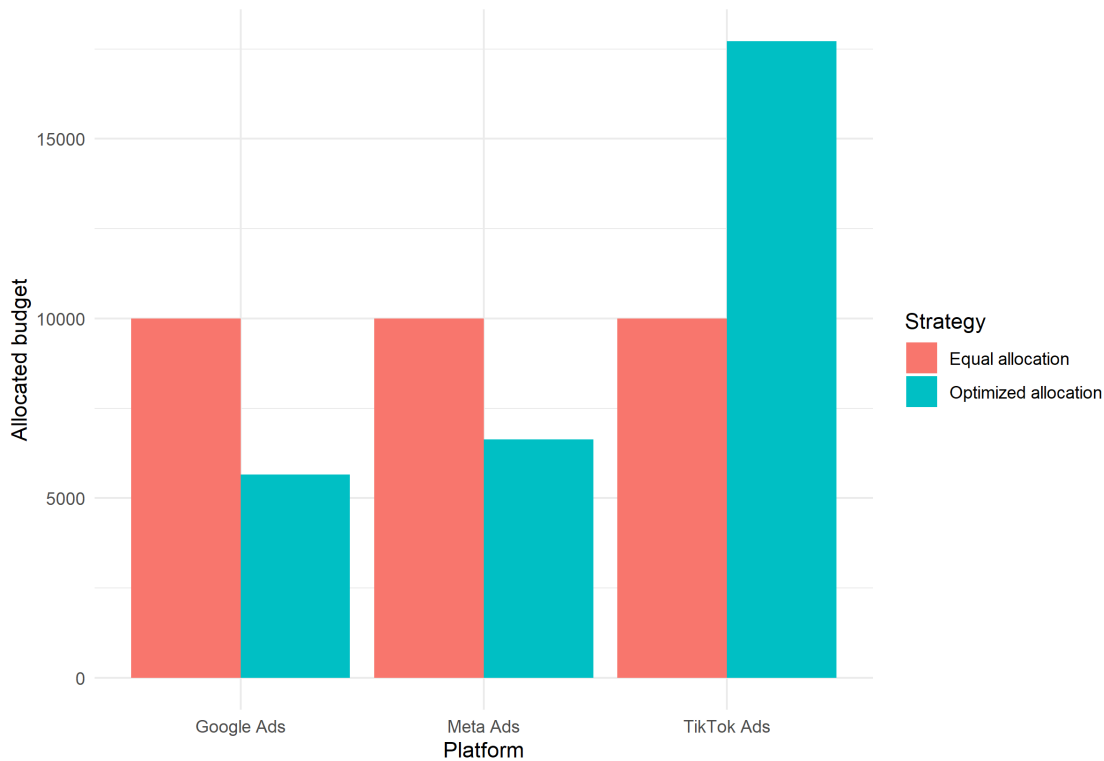


Figure 5: Comparison of equal and optimized budget allocation across platforms

The optimization results show a clear shift in allocation toward TikTok Ads. Under equal allocation, each platform receives \$10,000. The optimized allocation assigns approximately \$5,652 to Google Ads, \$6,634 to Meta Ads, and \$17,714 to TikTok Ads, corresponding to shares of 19%, 22%, and 59%. This redistribution reflects differences in marginal return across platforms and aligns with Figure 5.

This result highlights the importance of marginal-return-based decision making. Allocating budget equally across platforms ignores differences in efficiency and return, whereas the optimized strategy reallocates resources toward the most productive opportunities within the dataset. As a result, predicted revenue increases from approximately \$110,178 to \$115,691. This reinforces that optimal allocation depends on relative efficiency rather than equal distribution, even when all platforms are available to advertisers.

4.8 Platform and Campaign-Type Interactions

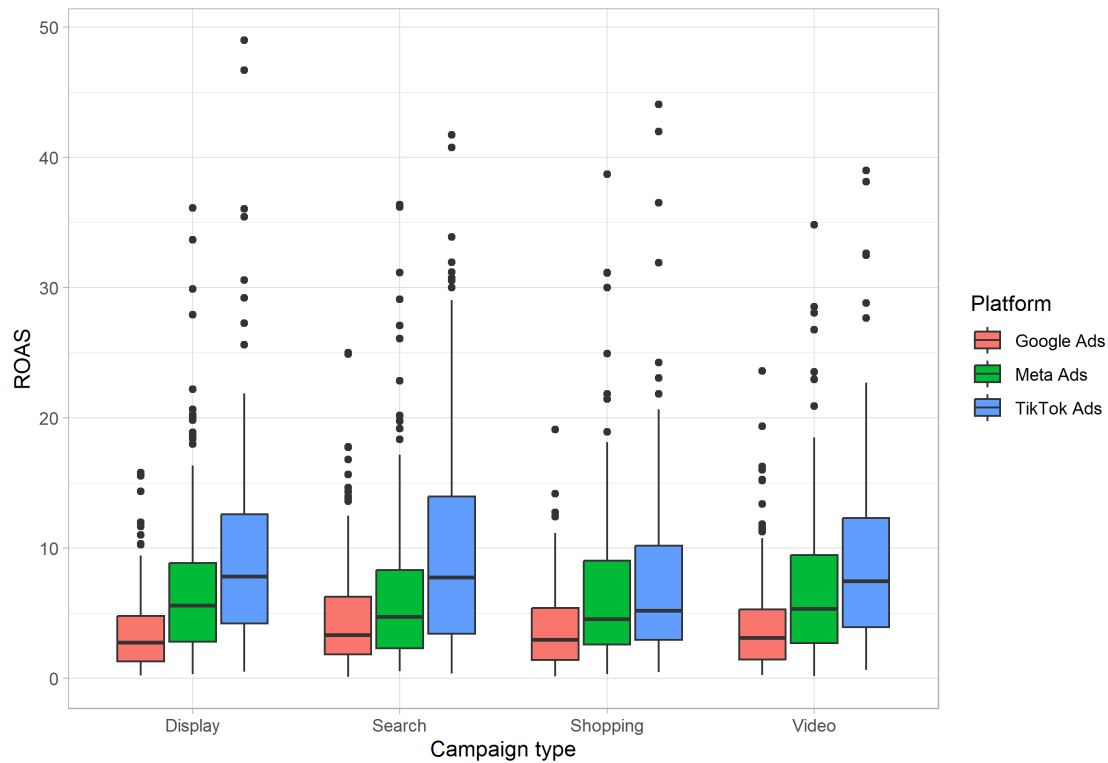


Figure 6: Interaction between platform and campaign type on ROAS

Platform performance varies across campaign types. TikTok Ads remains strong overall, but its advantage decreases for Search and Shopping campaigns. This indicates that platform effectiveness is not constant across campaign objectives and should be evaluated jointly with campaign type. Relying on a single average platform effect may therefore obscure important differences across campaign settings.

4.9 Joint Optimization Across Platform and Campaign Type

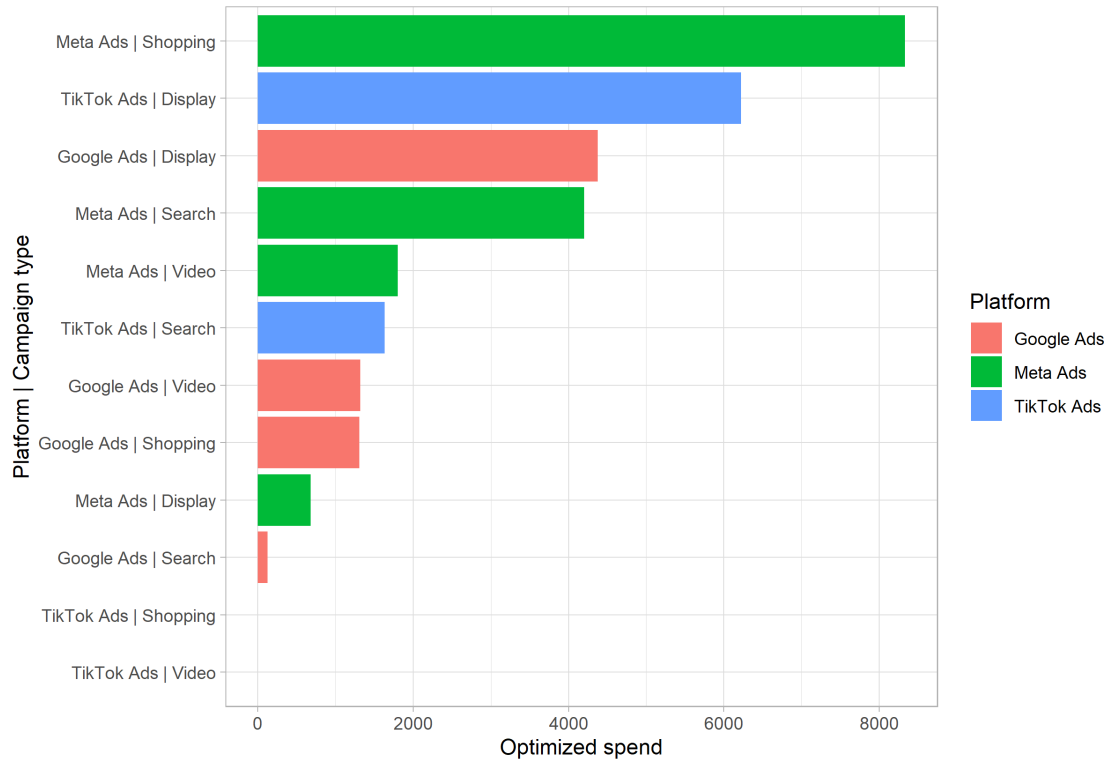


Figure 7: Optimized allocation across platform-by-campaign-type combinations

When optimization includes campaign type, budget allocation becomes more selective. Resources are concentrated in combinations with stronger estimated returns rather than being evenly distributed. Predicted revenue increases from approximately \$161,045 under equal allocation to \$179,547 under optimized allocation.

This demonstrates that incorporating campaign objectives into allocation decisions can substantially improve performance beyond platform-level optimization alone. It reinforces the idea that platform choice and campaign type should be considered jointly when making budget decisions.

4.10 Predictive Modeling Performance

The random forest model achieves lower prediction error than the linear regression model, with RMSE decreasing from 0.3772 to 0.2162. This improvement indicates that nonlinear relationships and interactions play an important role in predicting conversions. It also suggests that more flexible modeling approaches may better capture the complexity of digital advertising outcomes than simple linear models.

5 Conclusion

This study examined how platform choice, campaign type, and budget allocation shape digital advertising performance within a simulated campaign dataset. Across the analyses, platform emerged as the strongest observed source of variation. TikTok Ads generally exhibited higher engagement and return metrics, Google Ads appeared more cost-intensive, and Meta Ads occupied a middle position. However, these differences should be interpreted as patterns within the simulated data rather than as definitive statements about real-world platforms.

Campaign type contributed a second layer of variation, although its standalone effects were more modest. While Search campaigns showed slightly stronger average performance, there was substantial overlap across campaign types. The interaction analysis demonstrated that platform effectiveness is not constant across campaign objectives, reinforcing the importance of evaluating platform and campaign type jointly.

The regression analysis provided clear evidence of diminishing returns to advertising spend. This finding has direct implications for budgeting, as it suggests that increasing spend on a single platform does not lead to proportional increases in revenue. The optimization results translated this insight into actionable allocation strategies, showing that marginal-return-based allocation outperforms equal allocation. The gains were even larger when platform and campaign type were modeled jointly.

The predictive modeling results further demonstrated that campaign performance involves nonlinear relationships and interactions. The random forest model substantially improved prediction accuracy relative to a linear benchmark, indicating that flexible modeling approaches may better capture the complexity of digital advertising outcomes.

A key limitation of this study is the use of simulated data. While the dataset appears to reflect realistic relationships, the underlying data-generating process is not fully specified. As a result, platform differences may partially reflect assumptions built into the simulation. Therefore, the findings should be interpreted as demonstrating the usefulness of the analytical framework rather than providing externally valid conclusions.

Future work should apply this framework to real-world campaign data, incorporate temporal dynamics, and extend the optimization approach to account for uncertainty and multiple objectives, such as balancing revenue, conversions, and cost efficiency.

5.1 Appendix G. Additional Model Outputs

Table 5: Model fit statistics for the revenue regression

Metric	Value
R-squared	0.4389
Adjusted R-squared	0.4383
Sigma	0.9333
AIC	4864.8060
BIC	4886.7881

Table 6: Predictive performance on the test set

Model	RMSE
Linear regression	0.3772
Random forest	0.2162

References

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